



IT3060 – Human Computer Interaction

BSc (Hons) in Information Technology – Year 3

Lab Exercise 02

User Research & Requirements Gathering for the FitFlow App Redesign

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Activity 1: Organize Interview / Survey / Observation Data

Building on the research plan from Lab Exercise 01, all raw data collected for the FitFlow redesign — interview transcripts (P1–P5), the user survey, moderated usability-testing notes from the three rounds described in the case study, in-app analytics exports, and public app store reviews — was gathered into a single organized dataset before analysis began.

1.1 Raw Data Inventory

Data Source	Format	Volume / Sample	Representative Example
Interview transcripts	Qualitative (audio + notes)	5 sessions (P1–P5)	P4 (trainer): "I'd want to see any injury flags before trusting the AI plan."
Survey responses	Quantitative + open text	n = 42 respondents	68% rated meal logging 2/5 or lower on ease-of-use.
Usability testing notes	Mixed (task metrics + notes)	3 moderated rounds (per case study)	Round 1: users hesitated at AI plan screen, unsure how to adjust it.
App analytics reports	Quantitative (in-app events)	Rolling 90-day export	42% drop-off on the meal-logging screen before saving an entry.
App store reviews	Qualitative (public text)	35 recent reviews (iOS + Android)	"Feels like a generic plan, not made for me" — 3.0★ review.

1.2 Data Cleaning

- Removed duplicate survey submissions and near-identical app store reviews (e.g., repeated one-line reviews from the same account).
- Discarded irrelevant entries not related to the FitFlow experience (e.g., billing/refund complaints unrelated to usability, spam reviews).
- Standardised participant/reviewer references (P1–P5 for interviewees, R-## for survey respondents, UT-## for usability test sessions) so items can be cross-referenced across sources.
- Corrected obvious transcription typos in interview notes and trimmed filler speech ("um", "like") from quotes before tagging.

1.3 Categorization and Storage

Cleaned items were logged as rows in a shared spreadsheet (mirroring a Notion/Miro-style database) with the following columns: Data ID, Source, Participant/Reviewer, Demographic, Topic Tag(s), Quote/Observation, Sentiment, and Linked Theme (added during Activity 2). Topic tags used were: Onboarding, Workout Experience, AI Personalization, Nutrition Tracking, Motivation, Social Features, and Privacy. A sample of the categorized data is shown below; the full spreadsheet contains all 5 interviews, 42 survey responses, 3 rounds of usability notes, the analytics export, and 35 app store reviews.

ID	Source	Participant / Ref	Demographic / Context	Topic Tag	Quote / Observation
D-001	Interview	P1	Regular user, 3 mths on app	Nutrition Tracking	"Logging a snack takes about six taps — I usually give up."
D-002	Interview	P2	Power user, trains 5x/wk	Motivation / Social	"The leaderboard is what keeps me coming back."
D-003	Interview	P3	Premium subscriber, 1 yr	AI Personalization	"I don't know why it picked these targets for me."
D-004	Interview	P4	Fitness trainer	AI Trust / Safety	"Clients could get hurt if the AI doesn't know about an old injury."
D-005	Interview	P5	New user, week 1	Onboarding	"I joined a challenge without meaning to — wasn't sure what I'd agreed to."
D-006	Survey (open-ended)	R-14	18–24, casual user	Nutrition Tracking	"Barcode scanner is buried, I never use it."
D-007	Survey (open-ended)	R-27	35–44, premium	AI Personalization	"Would trust the AI more if it explained its choices."
D-008	Usability test (Round 1)	UT-03	Task: set up AI plan	AI Personalization	Hesitated 14s on plan screen; asked "can I change this later?"
D-009	App analytics	—	Aggregate (all users)	Nutrition Tracking	42% abandon meal-logging flow before saving.
D-010	App store review	iOS reviewer	Public review, 2★	Onboarding	"Confusing at first — took me a week to find the community tab."
D-011	App store review	Android reviewer	Public review, 3★	Motivation	"Motivating when I remember to open it, but no reminders."
D-012	Usability test (Round 2)	UT-07	Task: join a challenge	Privacy / Social	Unsure if progress would be visible to strangers before joining.

Notable quotes and observations (e.g., D-001, D-003, D-005, D-009, D-012) were flagged with a "★ notable" marker in the sentiment column so they could be quickly located during thematic coding and affinity mapping in the following activities.

Activity 2: Perform Thematic Analysis

All organized data from Activity 1 was reviewed multiple times — first per source (interviews, then survey, then usability notes, analytics, and reviews), and then again across sources — to identify recurring patterns. Notable quotes and observations were coded, and codes were grouped into major themes with supporting sub-themes.

2.1 Coding Sample

Quote / Observation (Data ID)	Code	Theme
"Logging a snack takes about six taps — I usually give up." (D-001)	Slow manual logging	Nutrition Tracking Issues
"Barcode scanner is buried, I never use it." (D-006)	Low discoverability of shortcuts	Nutrition Tracking Issues
42% abandon meal-logging flow before saving. (D-009)	High task abandonment	Nutrition Tracking Issues
"I don't know why it picked these targets for me." (D-007)	No rationale for AI output	Lack of Personalization
Hesitated 14s on plan screen; "can I change this later?" (D-008)	Uncertainty about editing AI plan	Lack of Personalization
"Clients could get hurt if the AI doesn't know about an old injury." (D-004)	No safety/constraint input	Lack of Personalization
"The leaderboard is what keeps me coming back." (D-002)	Competition drives return visits	Poor Motivation (opportunity)
"Motivating when I remember to open it, but no reminders." (D-011)	No re-engagement nudges	Poor Motivation
"I joined a challenge without meaning to." (D-005)	Unclear consent/visibility on joining	Social & Privacy Confusion
Unsure if progress would be visible to strangers before joining. (D-012)	No visibility settings shown upfront	Social & Privacy Confusion
"Confusing at first — took me a week to find the community tab." (D-010)	Poor discoverability of features	Onboarding & Navigation Issues

2.2 Theme Summary

Theme	Sub-themes	Frequency / Strength	Supporting Evidence (Data IDs)	Insight Summary
Lack of Personalization & AI Trust	No rationale shown; no override control; no safety/injury input	High — raised independently by 4 of 5 interviewees and echoed in 2 survey open-text answers and usability Round 1	D-003, D-004, D-007, D-008	Users are willing to follow AI guidance, but only if they understand and can adjust it — trust is currently undermined by a lack of transparency and control.
Nutrition Tracking Issues	Slow manual entry; low use of barcode scan; high abandonment	High — supported by qualitative complaints (D-001, D-006) and quantitative analytics (42% drop-off, D-009)	D-001, D-006, D-009	Meal logging is the single biggest daily-use friction point and likely contributes directly to disengagement, since it is a repeated, high-frequency task.
Poor Motivation	No reminders/streak nudges; motivation drops without social competition	Medium — present in 2 interviews and 3 app store reviews	D-002, D-011	Social/competitive elements (leaderboards) are a strong motivator already, but the app does not proactively re-engage users who lapse.
Onboarding & Navigation Issues	Confusing tab structure; features hard to discover in first sessions	Medium — raised by new user P5 and 2 app store reviews	D-005, D-010	First-time experience does not clearly orient users to the three core areas (workout, nutrition, social), risking early churn.
Social & Privacy Confusion	Unclear visibility settings; accidental participation in public challenges	Medium-High — raised in interview, usability testing, and implicit in survey privacy-trust score	D-005, D-012	Users want social motivation but are held back by not understanding who can see their activity, which undermines confidence in engaging with community features.

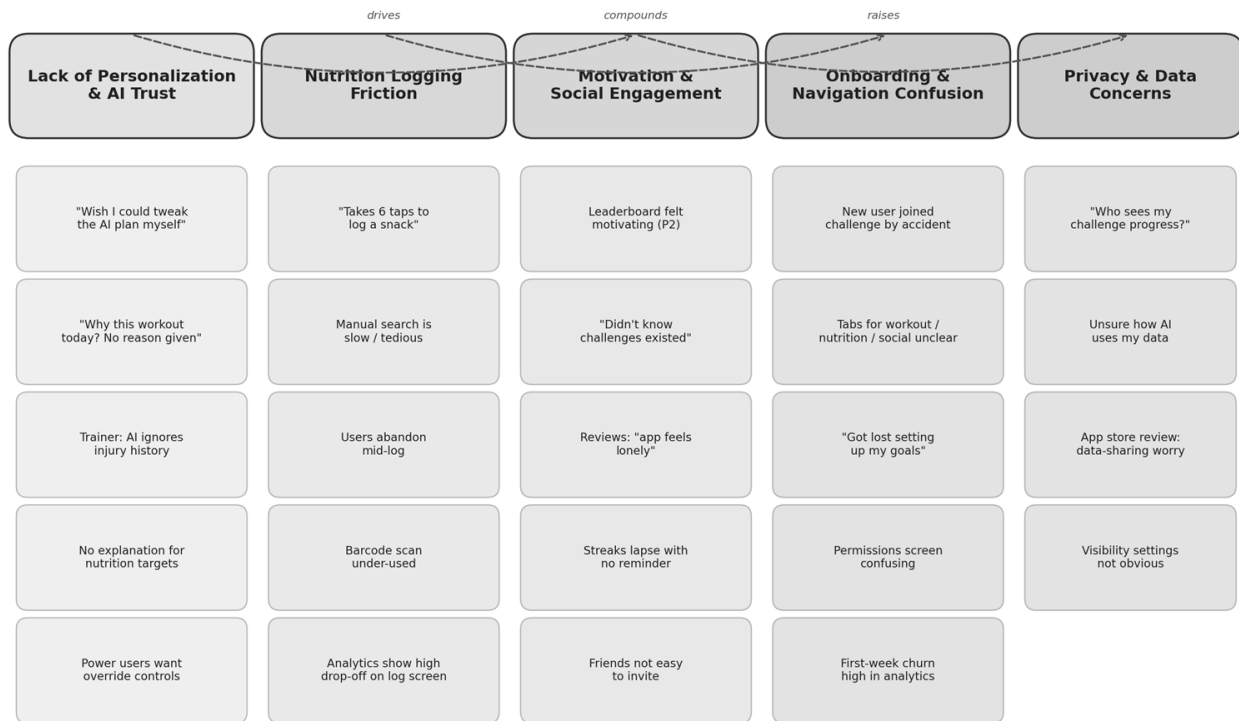
2.3 Validation

Themes were validated by re-reading the full dataset a second time against the theme list to check no significant recurring pattern was missed, and by cross-referencing each qualitative theme against a corresponding quantitative signal (e.g., the "Nutrition Tracking Issues" theme is supported both by direct complaints and by the 42% analytics drop-off figure). Two themes that initially looked separate — "confusing AI screen" and "no way to override AI" — were merged into the single theme "Lack of Personalization & AI Trust" since they shared the same underlying root cause. This validation step also matches the direction of the case study's own usability testing, which separately identified "more user control over AI suggestions" and "simplifying social navigation" as required improvements.

Activity 3: Create an Affinity Diagram

Each coded insight, user quote, and observation from Activity 2 was written onto an individual digital sticky note (simulating a Miro/FigJam-style board) and grouped into natural clusters. Sticky notes were sorted purely by content similarity first, then labelled once a clear pattern emerged in each group.

3.1 Affinity Diagram



Affinity Diagram — FitFlow User Research (digital sticky-note clusters)

Figure 1: Digital affinity diagram — sticky notes clustered into five theme groups, with dashed arrows showing identified relationships between clusters.

3.2 Cluster Relationships

- "Lack of Personalization & AI Trust" drives "Motivation & Social Engagement" concerns — users who don't trust the AI plan are less likely to stay engaged long enough to reach social features.
- "Nutrition Tracking Friction" compounds "Onboarding & Navigation Confusion" — new users who struggle with logging early on are more likely to abandon the app before habits form.
- "Motivation & Social Engagement" raises "Privacy & Data Concerns" — the more the app encourages social sharing, the more users need clarity on visibility and data use.

3.3 Key Findings from the Affinity Diagram

- Personalization and trust issues sit at the centre of the diagram because they connect to almost every other cluster — fixing AI transparency is likely to have the widest positive impact.
- Nutrition logging friction is a self-contained, high-frequency problem that can be solved with a focused UI fix (fewer taps, smarter defaults) independent of the other clusters.
- Social and privacy concerns are tightly linked — any redesign of community features must ship together with clearer visibility controls, not as separate work.

- Onboarding issues are a gateway problem: several other clusters (nutrition friction, privacy confusion) are first experienced during onboarding, so improving the first session has compounding benefits.

Activity 4: Develop Three Personas and Empathy Maps

Three primary user segments emerged from the research data and were developed into personas, each supported directly by evidence from the interviews, survey, and usability testing described in Activities 1–3.

Persona: Saantha Silva — "The Busy Professional"

Attribute	Details
Demographics	Age 32, marketing manager, lives in Colombo, exercises around a full-time job; profile closely matches interview participant P1 and premium subscriber P3.
Goals	Get an effective workout with minimal decision-making; keep nutrition roughly on track without spending much time logging; feel the plan is actually suited to her, not generic.
Behaviours	Opens FitFlow briefly during lunch breaks or early morning; relies on the AI-suggested plan rather than browsing a full exercise library; frequently forgets or skips logging meals when the day gets busy.
Frustrations	Meal logging takes too many taps to fit into a short break; doesn't understand why the AI chose certain nutrition targets, which makes her question whether to follow them; feels overwhelmed by notifications that aren't personalised to her schedule.
Supporting Research Evidence	"I don't know why it picked these targets for me" (P3, D-003); "Logging a snack takes about six taps — I usually give up" (P1, D-001).

Empathy Map

Says	Thinks
"I just want something quick that works — I don't have time to figure out the app."	"Is this plan actually right for me, or the same thing everyone gets?"
Does	Feels
Uses the app in short bursts (5–10 min); follows AI plans by default; skips meal logging on busy days.	Mildly frustrated by friction; slightly distrustful of unexplained AI choices; guilty when she skips logging.
Pains	Gains
Nutrition logging interrupts her day; no explanation for AI recommendations reduces confidence.	A plan that clearly adapts to her schedule and explains itself would save her time and build trust.

Persona: Anada Fernando — "The Fitness Enthusiast"

Attribute	Details
Demographics	Age 26, trains 5x a week, gym-goer with strong interest in performance data; profile matches interview participant P2 (power user).
Goals	Fine-tune his own training rather than fully defer to the AI; track detailed performance metrics over time; compete socially with friends to stay motivated.
Behaviours	Opens FitFlow daily; checks the community leaderboard regularly; attempts to manually adjust exercises or intensity within the AI-generated plan.
Frustrations	Cannot fully override or fine-tune AI-suggested intensity; wants more advanced analytics than currently shown; inviting friends to a challenge isn't straightforward.
Supporting Research Evidence	"The leaderboard is what keeps me coming back" (P2, D-002); wanted the ability to fully override AI-suggested intensity (Lab 1 findings, P2).

Empathy Map

Says	Thinks
"I know my body better than an algorithm does — let me tweak it."	"This app could be great if it just got out of my way when I want control."
Does	Feels
Trains daily; tries to manually adjust AI plans; checks leaderboards and challenge standings often.	Energised by competition; mildly annoyed when the app restricts customisation.
Pains	Gains
Limited override controls; analytics not detailed enough for a serious trainee.	Full manual control over AI suggestions plus deeper analytics would keep him engaged long-term and likely convert him into an advocate.

Persona: Sakuni Fernando — "The New Parent"

Attribute	Details
Demographics	Age 34, new mother returning to fitness after childbirth, limited and unpredictable free time; profile matches new/first-time user P5.
Goals	Fit in short, flexible workouts around unpredictable nap schedules; get simple, gentle nutrition guidance; stay motivated without feeling exposed or judged socially.
Behaviours	Uses the app in short, unplanned bursts; unsure of onboarding steps; cautious about anything shared publicly.
Frustrations	Onboarding flow was confusing and she was unsure what each tab did; accidentally joined a public challenge without understanding the privacy implications; wants reassurance before sharing anything.
Supporting Research Evidence	"I joined a challenge without meaning to — wasn't sure what I'd agreed to" (P5, D-005); unsure if progress would be visible to strangers before joining (usability Round 2, D-012).

Empathy Map

Says	Thinks
"I don't want strangers seeing my progress — I just want to ease back into things quietly."	"I'm not sure what half these buttons do yet."

Does	Feels
Opens the app in short, irregular sessions; hesitates before tapping into community features; re-checks settings after joining something by accident.	A little anxious about privacy; hopeful but easily discouraged if the first experience is confusing.

Pains	Gains
Unclear onboarding; no visibility confirmation before joining social features.	A guided first-run walkthrough and an explicit privacy choice before any social action would make her feel safe and in control.

All three personas were reviewed against the full dataset (Activities 1–3) to confirm they accurately represent distinct, recurring patterns in the user base rather than one-off opinions, and each is traceable to at least two independent data points.

Activity 5: Write User Stories

Twelve user stories were written using insights from the personas (Activity 4), themes (Activity 2), and affinity diagram (Activity 3), directly addressing current pain points and the new feature ideas emphasised in the case study — AI-personalised workouts, social/community features, and nutrition tracking. Each story follows the standard format and is prioritised by user impact and business goals (retention, premium conversion, and store ratings, as referenced in the case study).

ID	Persona	User Story	Priority	Linked Theme
US-01	Busy Professional	As a busy professional, I want the AI to show a short reason for each workout it picks, so that I can trust and follow the plan confidently.	High	Lack of Personalization & AI Trust
US-02	Busy Professional	As a busy professional, I want to log a meal in three taps or fewer using my recent or favourite items, so that logging doesn't interrupt my day.	High	Nutrition Tracking Issues
US-03	Fitness Enthusiast	As a fitness enthusiast, I want to manually adjust the intensity or swap exercises in my AI-generated plan, so that it matches my own training goals.	High	Lack of Personalization & AI Trust
US-04	Fitness Enthusiast	As a fitness enthusiast, I want to see detailed performance analytics over time, so that I can track my progress precisely.	Medium	Poor Motivation
US-05	Fitness Enthusiast	As a fitness enthusiast, I want to invite friends directly to a challenge from my profile, so that we can compete together easily.	Medium	Poor Motivation
US-06	New Parent	As a new parent, I want a short, flexible workout option (10–15 minutes), so that I can exercise during naps without feeling overwhelmed.	High	Onboarding & Navigation Issues
US-07	New Parent	As a new parent, I want to set who can see my progress before I join any community challenge, so that I feel safe participating.	High	Social & Privacy Confusion
US-08	New Parent	As a new parent, I want a simple guided walkthrough of the workout, nutrition, and social tabs on first launch, so that I don't get lost early on.	High	Onboarding & Navigation Issues
US-09	All personas	As a user, I want a reminder before my activity streak lapses, so that I stay motivated to continue.	Medium	Poor Motivation
US-10	Busy Professional / Premium subscriber	As a user, I want to see a short rationale next to AI nutrition recommendations, so that I understand and trust the targets given to me.	High	Lack of Personalization & AI Trust
US-11	Fitness Enthusiast / Trainer	As a user (or my trainer), I want to record injury or medical constraints, so that the AI avoids recommending unsafe exercises.	High	Lack of Personalization & AI Trust
US-12	New Parent / All personas	As a user, I want to control whether my challenge participation is private, friends-only, or public, so that my privacy is protected.	High	Social & Privacy Confusion

Prioritisation rationale: stories addressing AI transparency/trust (US-01, US-03, US-10, US-11) and privacy/onboarding (US-06, US-07, US-08, US-12) were rated High because they affect first impressions, safety,

and the trust needed for premium conversion. Analytics and social-invite stories (US-04, US-05, US-09) were rated Medium as they enhance engagement but are not blocking core adoption.

Activity 6: Create User Journey Maps

6.1 Current-State Journey

Stage	Actions	Thoughts	Emotions	Pain Points	Opportunities
Awareness	Sees an ad or hears about FitFlow from a friend; reads app store listing.	"Could this finally help me stay consistent?"	Hopeful, curious	Store reviews mention vague or generic-sounding claims.	Highlight AI personalisation and social proof clearly in store listing.
Download & Onboarding	Downloads app, creates account, grants permissions, sets basic goals.	"What does each of these tabs actually do?"	Slightly confused, uncertain	No guided walkthrough; permissions screen unclear (D-010).	Add a short guided first-run walkthrough (US-08).
First Workout / Meal Log	Tries the AI-generated plan; attempts to log first meal.	"Why did it give me this? Can I change it later?"	Cautious, mildly frustrated	No rationale for AI plan (D-008); meal logging takes ~6 taps (D-001).	Show AI rationale (US-01, US-10); simplify meal logging (US-02).
Regular Use (Weeks 2–4)	Logs workouts semi-regularly; occasionally checks community tab.	"This is fine, but it doesn't feel made for me."	Neutral, drifting	Generic-feeling plan; unaware challenges exist (D-011).	Personalise nutrition targets with rationale; surface challenges proactively.
Plateau / Frustration	Logging becomes inconsistent; misses days without reminders.	"I keep forgetting, and nothing nudges me back."	Frustrated, disengaging	No streak/re-engagement reminders (D-011); accidental challenge join eroded trust (D-005).	Add streak reminders (US-09); fix visibility controls (US-07, US-12).
Long-term Use / Advocacy	Some users churn quietly; others (like power users) stay for the leaderboard.	"I only stayed because of the leaderboard, not the AI."	Mixed — loyal minority, disengaged majority	Retention relies on one feature (social) rather than the whole experience.	Strengthen AI trust and nutrition ease so advocacy isn't dependent on one feature alone.

6.2 Future / Ideal-State Journey

Stage	Actions	Thoughts	Emotions	Solutions / Features Introduced
Awareness	Sees updated store listing highlighting explainable AI coaching and community challenges.	"This sounds like it actually adapts to me."	Hopeful, confident	Clear explanation of AI personalisation and social features (US-01, US-05).
Download & Onboarding	Completes a short guided walkthrough explaining workout, nutrition, and social tabs; sets a privacy preference upfront.	"Okay, I understand what each part does and I'm in control."	Oriented, reassured	Guided onboarding (US-08) and upfront privacy choice (US-07, US-12).
First Workout / Meal Log	Sees a short rationale for the AI plan; logs first meal in under 3 taps from recent items.	"That makes sense, and logging was quick."	Trusting, satisfied	AI rationale shown (US-01, US-10); fast meal logging (US-02).
Regular Use (Weeks 2–4)	Adjusts AI plan intensity directly; invites a friend to a challenge with clear visibility settings.	"This actually adapts to me, and I like competing with friends."	Engaged, motivated	Manual override (US-03); easy friend invites (US-05); analytics dashboard (US-04).
Plateau / Frustration (Reduced)	Gets a gentle reminder before a streak lapses; injury/constraint settings keep plans safe.	"It noticed I was slowing down and nudged me kindly."	Supported, less likely to disengage	Streak reminders (US-09); safety constraints (US-11).
Long-term Use / Advocacy	Continues using FitFlow for both personalised coaching and community challenges; recommends it to others.	"It feels made for me, not just another fitness app."	Delighted, loyal	Combined trust in AI + social engagement drives the retention and rating gains referenced in the case study (35% retention increase, 4.5★ average).

6.3 Emotional Arc — Current vs. Future State

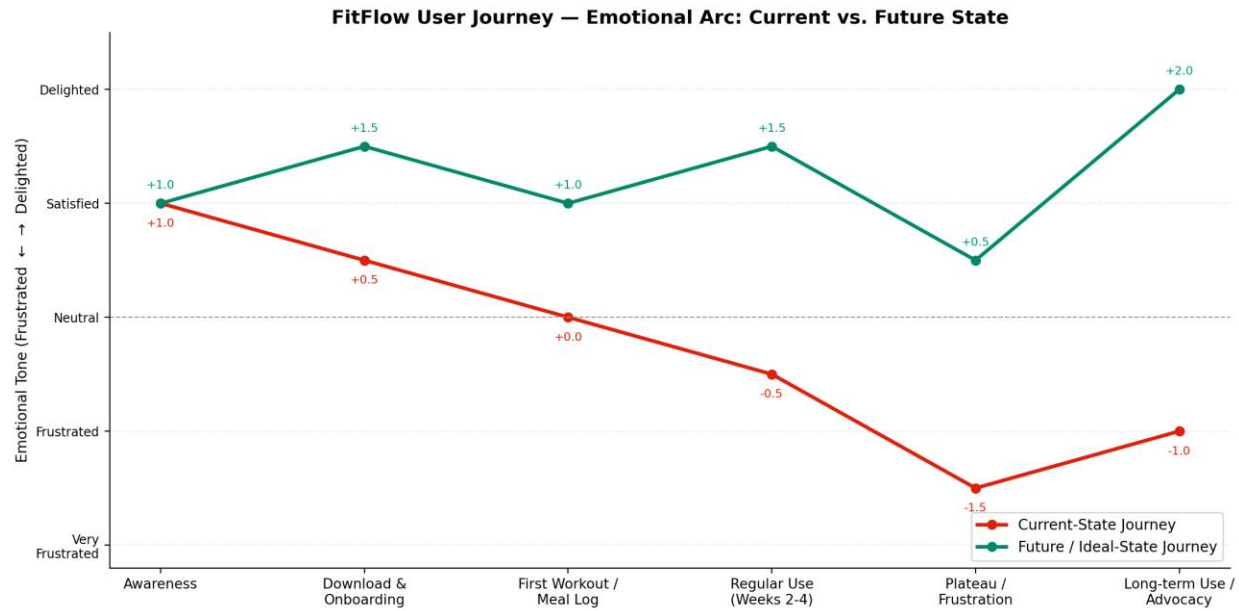


Figure 2: Emotional tone across the FitFlow user journey, comparing the current-state experience against the redesigned future-state experience.

The future-state journey removes the sharp dip at the "Plateau / Frustration" stage seen today and ends on a clearly positive note, consistent with the case study's reported outcomes after redesign — SUS score rising from 68 to 87 across three testing rounds, a 35% improvement in retention, and an average rating increase to 4.5 stars.

Activity 7: Generate How Might We (HMW) Statements

Key pain points and opportunities from the thematic analysis, affinity diagram, and journey maps were reframed into optimistic, open-ended HMW statements to inspire ideation. Five statements were generated, prioritized around the three areas the case study highlights as central to the redesign: personalization, social engagement, and nutrition tracking.

ID	Focus Area	How Might We Statement
HMW-1	Personalization / AI Trust	How might we help users understand and adjust AI-generated workout and nutrition plans so they feel personalised and trustworthy rather than generic?
HMW-2	Nutrition Tracking	How might we make logging a meal fast enough to fit into a busy day, so it stops feeling like a chore users give up on?
HMW-3	Social Engagement	How might we help users discover and confidently join community challenges, so social features become a strength rather than a source of confusion?
HMW-4	Motivation	How might we keep users motivated during a plateau, so they don't quietly disengage from FitFlow?
HMW-5	Onboarding	How might we make a new user's first session clear enough that they understand every core feature (AI plans, nutrition, social) without feeling lost?

Prioritization with the team ranked HMW-1 (Personalization/AI Trust) as the top priority, since it connects to the widest range of pain points identified in the affinity diagram (Activity 3), followed by HMW-2 (Nutrition Tracking) due to its high frequency and quantitative support from analytics, and HMW-3 (Social Engagement), which directly supports the case study's community and retention goals. HMW-4 and HMW-5 were retained as supporting statements that reinforce the top three.

Activity 8: Functional and Non-Functional Requirements with Traceability

Insights from the themes (Activity 2), affinity diagram (Activity 3), personas (Activity 4), user stories (Activity 5), journey maps (Activity 6), and HMW statements (Activity 7) were translated into a consolidated set of functional and non-functional requirements, each prioritised High, Medium, or Low.

8.1 Requirements List

ID	Requirement	Type	Priority
FR-01	The system shall display a short rationale whenever an AI-generated workout or nutrition recommendation is shown.	Functional	High
FR-02	The system shall allow users to manually edit, swap, or override any exercise or intensity level within an AI-generated plan.	Functional	High
FR-03	The system shall allow users to log a meal in three taps or fewer using recent, frequent, or favourite items.	Functional	High
FR-04	The system shall provide barcode scanning as an alternative nutrition-logging entry point.	Functional	Medium
FR-05	The system shall require users to set a visibility level (private / friends / public) before joining or sharing any community challenge.	Functional	High
FR-06	The system shall send a reminder notification before an active streak lapses.	Functional	Medium
FR-07	The system shall provide a guided onboarding walkthrough explaining the workout, nutrition, and social tabs on first launch.	Functional	High
FR-08	The system shall allow users (or their trainer) to record injury or medical constraints that the AI must not violate when generating plans.	Functional	High
FR-09	The system shall allow users to invite friends directly to a community challenge from their profile.	Functional	Medium
FR-10	The system shall display progress analytics and trends over a selectable time period (week / month / year).	Functional	Medium
NFR-01	AI-generated plan recommendations shall load within 2 seconds on a standard mobile connection.	Non-functional (Performance)	High
NFR-02	On-device personalization shall be processed locally (e.g., via TensorFlow Lite) to minimize transmission of sensitive personal data.	Non-functional (Privacy/Security)	High
NFR-03	The guided onboarding flow shall achieve at least 90% completion without drop-off, verified through usability testing.	Non-functional (Usability)	Medium
NFR-04	The application shall comply with WCAG 2.1 AA accessibility guidelines.	Non-functional (Accessibility)	High
NFR-05	The redesigned application shall maintain a System Usability Scale (SUS) score of at least 80 across usability testing rounds.	Non-functional (Usability)	High
NFR-06	All user health, nutrition, and location-related data shall be encrypted both in transit and at rest.	Non-functional (Security)	High

8.2 Traceability Matrix

Each requirement is linked back to the theme, persona(s), user story, and HMW statement that generated it, ensuring every requirement can be traced to genuine research evidence rather than assumption.

Req. ID	Related Theme(s)	Related Persona(s)	Related User Story ID(s)	Related HMW
FR-01	Lack of Personalization & AI Trust	Busy Professional, Fitness Enthusiast	US-01, US-10	HMW-1
FR-02	Lack of Personalization & AI Trust	Fitness Enthusiast	US-03	HMW-1
FR-03	Nutrition Tracking Issues	Busy Professional	US-02	HMW-2
FR-04	Nutrition Tracking Issues	Busy Professional	US-02	HMW-2
FR-05	Social & Privacy Confusion	New Parent	US-07, US-12	HMW-3
FR-06	Poor Motivation	All personas	US-09	HMW-4
FR-07	Onboarding & Navigation Issues	New Parent	US-08	HMW-5
FR-08	Lack of Personalization & AI Trust	Fitness Enthusiast (trainer input)	US-11	HMW-1
FR-09	Poor Motivation	Fitness Enthusiast	US-05	HMW-3
FR-10	Poor Motivation	Fitness Enthusiast	US-04	HMW-4
NFR-01	Lack of Personalization & AI Trust	All personas	US-01, US-03	HMW-1
NFR-02	Privacy / Data Concerns (case study tech stack)	All personas	US-12	HMW-1, HMW-3
NFR-03	Onboarding & Navigation Issues	New Parent	US-08	HMW-5
NFR-04	Onboarding & Navigation Issues	All personas	US-08	HMW-5
NFR-05	All themes (overall usability)	All personas	All user stories	HMW-1 – HMW-5
NFR-06	Social & Privacy Confusion	New Parent	US-07, US-12	HMW-3

8.3 Review Notes

- Clarity: each requirement uses a single "shall" statement to avoid combining multiple behaviors in one line.
- Feasibility: requirements referencing on-device AI (NFR-02) and accessibility (NFR-04) align with the technology stack described in the case study (TensorFlow Lite for on-device personalization, plus a privacy-focused architecture), so they are realistic given the chosen tools.
- Completeness: every theme from Activity 2 and every HMW statement from Activity 7 has at least one corresponding requirement, confirming no major research insight was left unaddressed.

Submission Checklist

- Activity 1: Organized data samples — raw data inventory and categorized sample table (Section 1).
- Activity 2: Coding/theme tables — coding sample and theme summary table (Section 2).
- Activity 3: Affinity diagram screenshot — Figure 1 (Section 3).
- Activity 4: Three personas and empathy maps (Section 4).
- Activity 5: 12 prioritized user stories (Section 5).
- Activity 6: Current-state and future-state journey maps plus emotional arc chart — Figure 2 (Section 6).
- Activity 7: Five HMW statements with prioritization (Section 7).
- Activity 8: Requirements list and traceability matrix (Section 8).